

AI-Assisted Learning and the Reversal of Long-Run Skill Degradation

wbgradost

<https://github.com/wbgradost/ai-project>

Topic presentation, October 2026

1 The question and the track

Can AI assistance raise long-run skill when it not only substitutes for human effort in current production, but also makes the remaining effort more effective at producing skill? I study this question under **Track A** by changing one assumption in the endogenous-skill model of [Aouad et al. \(2026\)](#). Their Proposition 3.5 (p. 8) establishes a weak downward shift of the stationary skill distribution as AI assistance rises. My conjecture is deliberately narrower: sufficiently strong AI–effort complementarity in learning reverses that comparison over an initial range of assistance, but effort displacement eventually dominates. The objective is an exact two-state characterization, not a universal claim that more AI always improves skill.

2 The baseline and the assumption I change

Worker problem. A myopic worker has skill $s > 0$, chooses effort $e \geq 0$, and receives AI assistance $a \geq 0$. Skill, effort, and AI are perfect substitutes in current production. With an increasing, concave production function p and linear effort cost γe , $\gamma > 0$, the worker solves

$$e^*(s, a) \in \arg \max_{e \geq 0} \{p(s + e + a) - \gamma e\}. \quad (1)$$

The Kuhn–Tucker conditions are

$$e^* \geq 0, \quad \gamma - p'(s + e^* + a) \geq 0, \quad e^* [\gamma - p'(s + e^* + a)] = 0. \quad (2)$$

Let x^* be the largest maximizer of $p(x) - \gamma x$. Under the paper’s boundedness condition, Proposition 2.1 gives

$$e^*(s, a) = (x^* - s - a)^+. \quad (3)$$

Thus, while effort is positive, an additional unit of AI displaces one unit of effort.

Baseline skill dynamics. The original model is a continuous-time birth–death chain. From state s_i , the upward rate is $\lambda(e^*(s_i, a))$, where λ is positive, increasing, differentiable, and weakly concave; the downward rate is the constant $\mu > 0$. Local balance gives

$$\frac{\pi_{i+1}(a)}{\pi_i(a)} = \frac{\lambda(e^*(s_i, a))}{\mu}. \quad (4)$$

Because (3) is decreasing in a and the baseline λ is increasing in effort, every adjacent stationary odds ratio falls with AI. Proposition 3.5 therefore shows that lower assistance first-order stochastically dominates higher assistance. Its Appendix A.4 proves this result by comparing normalized sums of products of the ratios in (4).

Changed assumption. I preserve (1), myopia, current production, and the downward rate, but replace the effort-only learning technology with

$$\lambda(e, a) = \lambda_0 + \kappa e(1 + \eta a), \quad \lambda_0 > 0, \kappa > 0, \eta \geq 0. \quad (5)$$

For each fixed a , (5) is positive, increasing, and linear (hence weakly concave) in effort. Its cross-partial is $\lambda_{ea} = \kappa\eta \geq 0$: AI raises the learning effectiveness of effort. When $\eta = 0$, (5) is an affine special case of the baseline. Since the worker remains myopic, (5) does not change the first-order conditions (2); it changes only the transition rate evaluated at optimal effort.

3 The result I expect

Consider two skill states $s_L < s_H$. Write $\lambda_L(a) = \lambda(e^*(s_L, a), a)$. Stationary flow balance and normalization are

$$\pi_L \lambda_L = \pi_H \mu, \quad \pi_L + \pi_H = 1, \quad \implies \quad \pi_H(a) = \frac{\lambda_L(a)}{\lambda_L(a) + \mu}. \quad (6)$$

Let $b = x^* - s_L > 0$. By (3) and (5),

$$\lambda_L(a) = \lambda_0 + \kappa(b - a)^+(1 + \eta a). \quad (7)$$

In the active-effort region $0 \leq a < b$,

$$\begin{aligned} \lambda_L(a) &= \lambda_0 + \kappa[b + (\eta b - 1)a - \eta a^2], \\ \frac{d\lambda_L}{da} &= \underbrace{\kappa\eta(b - a)}_{\text{learning complementarity}} - \underbrace{\kappa(1 + \eta a)}_{\text{effort crowd-out}} \\ &= \kappa[\eta(b - 2a) - 1], \end{aligned} \quad (8)$$

$$\frac{d\pi_H}{da} = \frac{\mu\kappa[\eta(b - 2a) - 1]}{[\lambda_L(a) + \mu]^2}. \quad (9)$$

The positive term in (8) is the direct increase in learning effectiveness; the negative term is the loss caused by effort crowd-out. This yields the candidate proposition. If $\eta b \leq 1$, the high-skill share is weakly decreasing, as in the baseline. If $\eta b > 1$, it increases up to

$$a^\dagger = \frac{\eta b - 1}{2\eta}, \quad (10)$$

then decreases for $a^\dagger < a < b$. At the second threshold $a = b$, effort reaches zero; thereafter $\lambda_L = \lambda_0$ and $\pi_H = \lambda_0/(\lambda_0 + \mu)$ is constant. For any $0 \leq a_\ell < a_h < b$, the exact pairwise condition is

$$\pi_H(a_h) > \pi_H(a_\ell) \iff \eta(b - a_h - a_\ell) > 1. \quad (11)$$

Equation (11) is the proposed local reversal of Proposition 3.5. It also makes the limitation explicit: if the higher assistance level crosses b , the complementarity term disappears and cannot sustain a global reversal.

4 Why this is not already solved

I reviewed the current arXiv version of [Aouad et al. \(2026\)](#), including Appendix A.4 and the convex-cost extension in Appendix D. Their skill transition remains a function of effort alone; the appendices do not introduce AI into the learning technology or derive (8)–(11). ArXiv currently lists only version 1; the authors also identify an EC 2026 conference version.

This project does *not* claim that AI–learning complementarity is new. Most directly, [Davies \(2026\)](#) studies a myopic student who may work or delegate to AI. His Theorem 4 lets AI quality raise the marginal product of effort and shows that higher quality can increase learning for high-skill students while reducing it for low-skill students. That model endogenizes course design and discrete delegation, whereas this project keeps the worker problem and stationary birth–death structure of [Aouad et al. \(2026\)](#) and puts complementarity directly in the transition rate. The added result is the closed-form stationary reversal condition and its assistance threshold. Recent experiments also distinguish productive AI use from effort-replacing use ([Shen and Tamkin, 2026](#); [Wu et al., 2026](#)); they motivate the mechanism but do not provide this stationary comparative static.

5 Plan and risks

Proof. I will prove the two-state proposition with all boundary and equality cases, then determine whether state-by-state transition inequalities yield a useful N -state stochastic-dominance result.

Verification. After the proposition is fixed, a symbolic script will verify (6)–(11), and a parameter-grid simulation will test sign changes, threshold crossings, and limiting cases; these are checks, not proofs. **Lean.** I will formalize the two-state stationary identity, the sign equivalence in (9), and the pairwise condition (11), including positivity assumptions and the piecewise effort boundary. These components are intentionally deferred until the economic statement has been reviewed.

The main mathematical risk is that the clean two-state ordering need not extend to N states when different effort thresholds are crossed. The fallback is to state exact sufficient conditions on every upward rate rather than claim a global reversal. The main economic risks are the maintained myopia assumption and the fact that, in two states, high-state effort affects output but not further learning. Finally, (5) is concave in effort for fixed AI but not jointly concave in (e, a) ; the paper will state the required domain-specific assumption explicitly.

References

- Ali Aouad, Thodoris Lykouris, and Huiying Zhong. Human–AI productivity paradoxes: Modeling the interplay of skill, effort, and AI assistance, 2026. arXiv preprint, version 1; conference version at the 27th ACM Conference on Economics and Computation (EC 2026).
- Benjamin Davies. Course design in the age of AI, 2026. Working paper, version 3, September 2026.
- Judy Hanwen Shen and Alex Tamkin. How AI impacts skill formation, 2026. Preprint, version 2.
- Shang Wu, Catarina G. Belem, Shuyuan Fu, Mark Steyvers, and Padhraic Smyth. How AI assistance affects human skill development: A study of learning with logic puzzles, 2026. Preprint; accepted at Human–AI Complementarity and Alignment (HCOMP) 2026.